# Autobahn: Efficient Autoregressive Video Generation Based on Semantic–Temporal Reduction

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***Abstract*—Video visual autoregressive (VAR) models have emerged as a promising approach for video generation. However, their next-scale prediction incurs substantial Transformer computation because the number of tokens rapidly increases at later scales. Token reduction has been explored to mitigate this cost, but existing methods are designed for already formed visual content and therefore assume redundancy among spatially or temporally neighboring tokens. This assumption is ill-suited to video VARs, which progressively construct visual content across decoding scales, leading to accuracy degradation. To address this limitation, we examine video VAR features and observe two complementary forms of redundancy that emerge during video VAR inference. First, the semantically complete reference image generated before video decoding is strongly correlated with subsequent video features, providing a stable semantic reference. Second, temporal redundancy in video VAR features follows an interleaved pattern distinct from the natural temporal structure of video data. Based on these observations, we propose Autobahn, a semantic–temporal reduction framework for efficient video VAR generation. Autobahn first identifies features aligned with the reference image and then captures temporal redundancy through position-aligned comparison between frame pairs. After identifying redundant features, Autobahn recovers skipped positions using the corresponding features from the previous scale by exploiting the inter-scale correlation of VAR generation. We further design the Autobahn hardware to accelerate identification, compaction into a dense sequence, and restoration of the original feature-map layout with low data-movement overhead. As a result, Autobahn achieves up to 2.54× speedup and improves generation quality by an average of 3.8 dB in PSNR compared to other accelerators.‘**

## I. Introduction

Visual autoregressive (VAR) models [5], [24], [25] have emerged as a promising generative paradigm for visual content by reformulating conventional next-token prediction as coarse-to-fine next-scale prediction. Unlike conventional autoregressive models that generate visual tokens sequentially [23], [27], [29], VARs predict all tokens within each scale in parallel and condition each finer scale on the previously generated coarse representation. This scale-wise generation substantially shortens the sequential inference path, reducing the computational cost of autoregressive generation. Recently, video VARs have extended this paradigm to video [15], enabling visual representations to be progressively refined across both spatial and temporal dimensions and providing an efficient framework for high-fidelity video generation.

However, scale-level parallelism does not eliminate the computational cost within each scale. In video VARs, the temporal length of a clip remains fixed, while the spatial token map grows rapidly at later scales. Consequently, the number of spatio-temporal tokens increases substantially as generation proceeds to later, higher-resolution scales [4], [11], thereby incurring significant Transformer computation.

To alleviate this cost, token reduction [1], [10], [12], [21], [32] has been widely studied in vision deep neural networks (DNNs). These methods require identifying unimportant or redundant tokens by computing pairwise similarities within a candidate search space, removing the selected tokens from Transformer computation, and recovering their positions using representative features. Consequently, their end-to-end effectiveness depends on both efficient and reliable token identification and faithful recovery.

However, existing methods are primarily designed for conventional vision DNNs such as ViTs [3], [7], [32], [33] and video ViTs [21], where visual content is already available. Through our analysis, we identify three key challenges in applying them to video VARs. First, the cost of token identification grows with the candidate search space, as each token is compared against multiple other tokens within the search space. Second, unlike conventional vision DNNs, video VARs progressively construct visual content across scales, making the spatial or temporal locality assumed by existing methods unreliable. Finally, existing recovery schemes use a shared representative feature to reconstruct multiple token positions, potentially discarding position-specific information and introducing distortion during visual generation.

To address these challenges, we analyze the video VAR inference pipeline and its feature characteristics. In video VARs, the reference image is generated before video generation, providing a semantically complete representation for token identification. Our attention-pattern analysis further shows that each frame query consistently attends strongly to spatially

aligned positions in the reference image. Together, these observations indicate that aligned reference-image positions can serve as predetermined candidates for identifying semantic redundancy.

However, the reference image alone cannot capture frame-dependent motion and temporal variation. Through attention- and feature-map analyses, we further observe that, unlike conventional video data, video VAR features exhibit temporally interleaved phases, with similar patterns recurring across non-adjacent frames. This motivates grouping adjacent frames into a Temporal Probing Window (TPW), allowing recurring phases to be evaluated jointly for reliable temporal-redundancy identification.

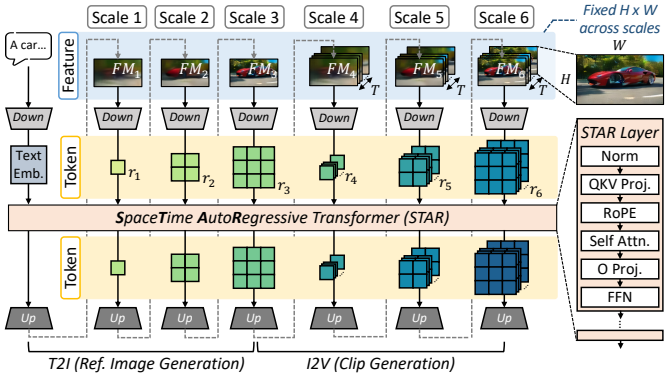
Finally, our feature analysis reveals strong similarity between aligned positions across adjacent scales due to the coarse-to-fine generation process. Thus, each removed token can be recovered directly from its aligned position at an adjacent scale, rather than reconstructing multiple positions from a single representative token, avoiding generation distortion.

Based on these observations, we propose Autobahn, an algorithm-hardware co-design for efficient autoregressive video generation. Autobahn first compares each target position with its aligned reference image feature and directly skips positions with high semantic similarity. Positions that remain unresolved are subsequently evaluated at the TPW level using position-aligned comparison with a shared temporal reference. The remaining update tokens are compacted in their original order and processed as a dense sequence in the Transformer computation. After execution, Autobahn reconstructs the full feature map by combining the updated outputs with the corresponding previous-scale features at skipped positions.

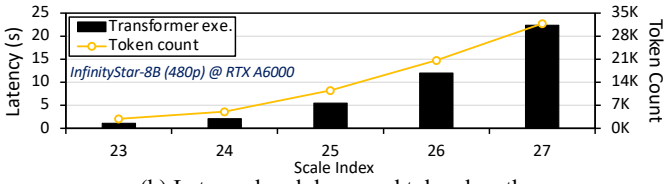
At the hardware level, Autobahn supports this flow using three dedicated components. The Identification Unit performs position-aligned reference comparison and TPW-level identification to generate a binary skip-bit table. The Compaction Unit converts the non-skipped positions into a dense sequence using prefix-sum-based address generation while preserving their original order. Finally, the Recovery Unit scans the skip-bit table and reconstructs the output layout by selecting either the next compacted Transformer output or the corresponding previous-scale feature. This order-preserving dataflow integrates output reordering and token recovery into a single streaming pass while keeping the main GEMM and attention datapath dense.

In summary, our main contributions are as follows:

- We propose Autobahn, a semantic-temporal token-reduction algorithm that restricts candidate comparison to aligned positions, captures temporal redundancy at the TPW level, and recovers skipped positions using aligned previous-scale features.
- We design dedicated hardware for token identification, prefix-sum-based compaction, and streaming recovery, mitigating representative index management and integrating output reordering and recovery into an order-preserving dataflow.




(a) Overview of Video VAR




(b) Latency breakdown and token length

Fig. 1. (a) Generation pipeline of video VAR. Video VAR generates a feature map $FM_k$ at each scale $k$ by passing down the token map $r_k$ in transformer blocks. Text-to-Image (T2I) and Image-to-Video (I2V) generate a reference image and clip, respectively. The time dimension is fixed across scales within the clip generation. (b) Latency breakdown and token length measured in InfinityStar-8B 480p.

- Our integrated algorithm and hardware achieve 2.54× speedup with enhanced generation quality of 3.8 dB on average PSNR over baseline accelerators.

## II. Background

### A. Preliminaries of Video Visual Autoregressive Model

Unlike conventional autoregressive image generation models, which generate one token per decoding step, VAR adopts next-scale prediction to decode multiple tokens in parallel and reduce the number of decoding stages. Motivated by the coarse-to-fine nature of visual content, VAR represents an image as a sequence of multi-scale token maps $\mathcal{R} = r_1, r_2, \ldots, r_K$, where each map $r_k \in \mathbb{Z}^{h_k \times w_k}$ is predicted conditioned on preceding lower-resolution maps. Consequently, early scales capture global structure, while later scales progressively refine local details.

More recently, space-time autoregressive (STAR) models extend this approach to video generation by decoupling spatial and temporal prediction, as shown in Fig. 1(a). At the Text-to-Image (T2I) stage, Video VAR first generates a single-frame clip ($T$=1) through coarse-to-fine image generation, which we refer to as the *reference image*. Then, at the Image-to-Video (I2V) stage, it generates subsequent fixed-length video clips ($T > 1$) according to the desired video length. During the generation of each video clip, the reference image serves as a conditioning context, allowing the video-frame tokens to attend to its features throughout the generation process.

Within each clip, the temporal length remains fixed while only the spatial resolution increases across decoding stages.

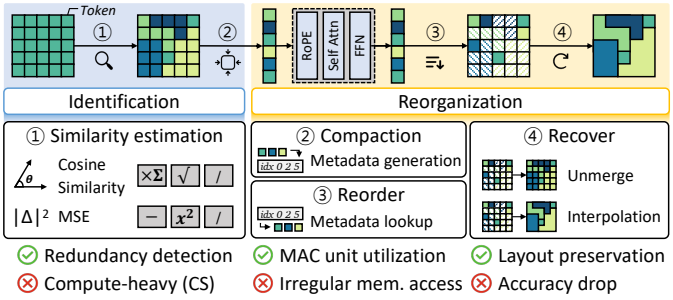
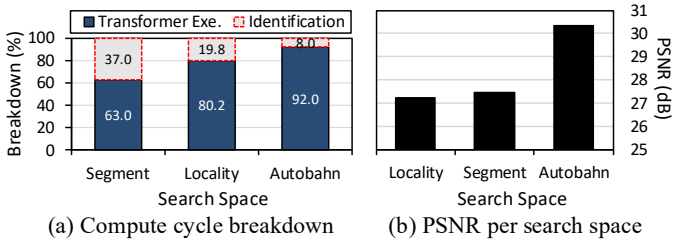



Fig. 2. Illustration of token reduction pipeline composed of token identification and reorganization.




(a) Compute cycle breakdown (b) PSNR per search space

Fig. 3. Identification stage limitation in video VAR. All methods reduced 70% of the original tokens. (a) Compute cycle breakdown of prior token reduction methods and our proposed method. (b) PSNR measured on various search spaces without recovery.

Although this design requires fewer decoding stages than video diffusion and conventional video autoregressive models, all tokens across the temporal and spatial dimensions of each scale must be processed in parallel within each Transformer block. The resulting token growth at higher spatial resolutions substantially increases computation and latency at later scales, as shown in Fig. 1(b).

### B. Token Reduction

Visual content often exhibits substantial token redundancy, as spatially or temporally correlated regions can be represented by tokens with similar semantics. This redundancy provides an opportunity to avoid repeatedly processing overlapping information. Accordingly, token reduction [30] has been widely studied as an effective approach for accelerating Transformer-based visual models. As shown in Fig. 2, token reduction consists of identification, compaction, reordering, and recovery. The identification stage detects redundant tokens by computing similarity scores within a candidate search space, while compaction packs the retained tokens into dense matrices for efficient Transformer computation. After computation, the outputs are reordered to their original positions, and removed tokens are optionally recovered from their representative tokens to preserve output quality.

However, token redundancy does not necessarily imply that tokens can be safely removed without quality loss. Tokens with similar representations may still contribute differently to video generation. Consequently, inaccurate redundancy identification or representative selection can degrade generation quality. This challenge becomes more severe under extreme sparsity, where a small set of retained tokens must preserve the information carried by a much larger set of removed tokens. In this regime, maintaining generation quality depends not only on accurately identifying redundant tokens, but also on effectively recovering the representations of removed tokens from their retained representatives.

To achieve low-latency, high-accuracy token reduction, it must satisfy two key requirements. First, the identification stage must accurately identify redundant tokens and select suitable representatives while minimizing the overhead of similarity scoring and candidate selection. Second, the recovery stage must effectively reconstruct the representations of removed tokens from their retained representatives.

## III. Motivation

### A. Limitation of Existing Token Reduction Method

To achieve effective token reduction in video VARs, we first examine existing methods based on the key requirements identified above. The effectiveness of the identification stage depends on the candidate search space, as it determines which tokens can be compared and selected as representatives. Likewise, the effectiveness of the recovery stage depends on the selected representative features and recovery mechanism, as they determine how accurately the reduced-token representations can be reconstructed. We first examine the candidate search space. A broader search space provides more opportunities to find a suitable representative but requires more similarity computations. Global search [1] uses the broadest search space by comparing each token against all other tokens. Although this provides broad candidate coverage, its high identification overhead, especially at later video VAR scales, makes it impractical for efficient inference.

Accordingly, recent work reduces the search space through structured candidate selection. Segment-based methods [21], [28], [35] partition the input into temporal segments, select a representative frame within each segment, and skip redundant tokens in the remaining frames based on their similarity to the representative. Locality-based methods [8], [19], [32] further restrict each token to candidates within a small spatial neighborhood and merge similar tokens into representative tokens. Although these approaches reduce the search space compared with global search, each token must still be compared against multiple candidates to identify a suitable representative. These repeated token-to-token comparisons incur high identification overhead for large spatiotemporal token maps. As shown in Fig. 3(a), despite reducing the candidate search space, their identification stages still account for 37% and 19% of the total execution cycles, respectively.

Beyond their identification overhead, existing methods do not fully account for how visual features are progressively formed during video VAR generation. They typically identify redundancy among spatially or temporally neighboring features extracted from already available visual content, as

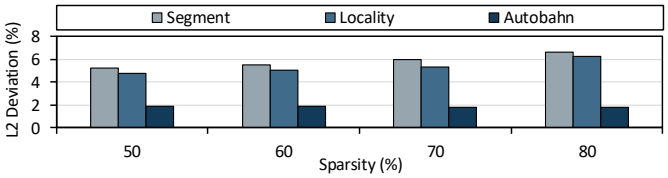



Fig. 4. L2 deviation of the feature map compared to the original feature map on various recovery methods.

in ViT or video ViT. In video VAR, however, the semantic structure of the visual representation is still being generated across decoding scales. As a result, spatially or temporally neighboring features may not be similar, making predefined locality a less reliable basis for selecting redundant tokens. To quantify this limitation, we measure the Peak Signal-to-Noise Ratio (PSNR) of the existing methods as shown in Fig. 3(b). In the figure, segment-based and locality-based approaches achieve 3.1 dB and 2.9 dB lower PSNR than Autobahn, respectively, demonstrating that their locality assumptions do not hold during progressive refinement and cause substantial accuracy degradation in video VAR generation.

In addition to accurate token identification, accurate recovery is essential for preserving generation quality under extreme sparsity [26], [34]. Segment-based methods recover skipped positions using features from a representative frame, thereby assigning the representative frame information to multiple temporal positions. Locality-based methods reconstruct merged positions from a shared representative-token output during unmerging, thereby propagating the same information across multiple spatial positions.

However, these recovery strategies may not be well suited to the generation process of video VAR. Using a single representative to recover multiple tokens assigns the same information to different spatial and temporal positions, suppressing their position-specific differences and distorting the spatial and temporal structure of the generated representation.

To quantify the importance of recovery, we measure the L2 deviation of the recovered feature maps from the non-reduced baseline. Fig. 4 shows that segment-based and locality-based recovery produce, on average, 3.2× and 3.0× larger deviations than Autobahn, respectively. Moreover, as sparsity increases from 50% to 80%, their deviations increase by 27% and 31%, whereas Autobahn maintains nearly unchanged L2 deviation. This result indicates that prior schemes become increasingly ineffective as higher sparsity leaves more missing positions to be recovered from a single representative feature.

Overall, these observations motivate a token-reduction method that leverages video VAR-specific redundancy to minimize candidate comparisons and token-identification errors while accurately recovering skipped positions.

### *B. Semantic and Temporal Candidate Spaces in Video VAR*

Since attention distributions indicate which tokens provide relevant information to each query, they offer a useful basis for identifying representative tokens. We therefore first analyze

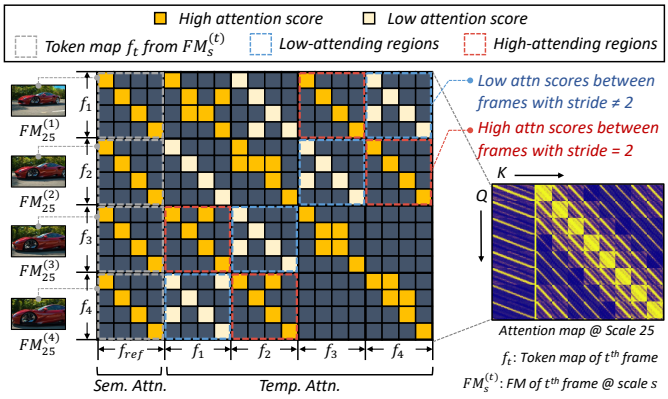


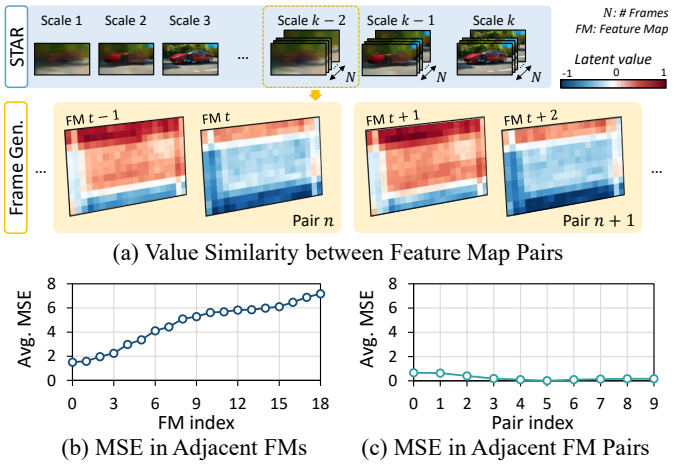

Fig. 5. Attention map during video clip generation in InfinityStar-8B at scale 25, Transformer layer 29, and attention head 31. Yellow and blue denote high and low attention scores, respectively.

the attention distribution of video VARs to identify effective candidate spaces for the identification stage, as shown in Fig. 5. Specifically, we profile the attention score at scale 25, Transformer layer 29, and across all attention heads. The figure presents the attention from video-frame queries to keys from both the reference image and the video frames. This reflects the native attention structure of video VAR, where the reference image serves as conditioning context for the generation of subsequent video clips as described in Section II-A.

On the left side of the figure, attention to the reference-image tokens exhibits multiple diagonal patterns repeated at intervals of $h_k w_k$. Each diagonal indicates that the tokens in each video frame attend strongly to reference-image tokens at corresponding spatial positions. Unlike the intermediate video-frame representations that are still being progressively generated, the reference image has already been completed through the video VAR inference pipeline and provides semantically established visual information. Therefore, each position-aligned reference image can serve as an accurate semantic candidate for the corresponding video-frame token. Notably, this position alignment establishes a one-to-one correspondence, eliminating the need for similarity search over the entire reference set.

In contrast, on the right side of Fig. 5, attention among video-frame tokens exhibits a structured temporal pattern. The main diagonal is the most prominent since tokens attend most strongly to spatially corresponding tokens within the same frame. However, attention is not confined to the same frame or its immediate temporal neighbors. Multiple off-diagonal bands recur at regular frame-token intervals, indicating structured correspondences across different temporal positions. This suggests that temporal correspondence is distributed across a recurring group of frames rather than being confined to an individual neighboring frame.

To further accurately capture the temporal behavior of video VAR features, we first examine the value distributions of consecutive feature maps, as shown in Fig. 6(a). Unlike adjacent




(a) Value Similarity between Feature Map Pairs

(b) MSE in Adjacent FMs

(c) MSE in Adjacent FM Pairs

Fig. 6. (a) Value heat map across 4 consecutive feature maps and paired adjacent feature maps within scale $k-2$. Average MSE was measured in (b) adjacent feature maps and (c) adjacent pairs. Lower MSE indicates greater similarity.

frames in natural video data, which generally exhibit strong temporal similarity, consecutively generated feature maps in video VAR frequently show inverted activation patterns. These inversions obscure their underlying similarity when the feature maps are compared independently. To quantify the similarity between the adjacent feature maps, we compute their average MSE, where a lower MSE indicates greater similarity. Accordingly, adjacent feature maps exhibit relatively high MSE, which generally increases across the frame sequence, as shown in Fig. 6(b).

However, we observe that the value distributions of multiple feature maps become highly similar when they are considered over specific temporal pairs. This suggests that grouping frames according to these recurring temporal patterns can provide a more reliable basis for selecting representatives and capturing temporal redundancy. We quantify this behavior in Fig. 6(c). The MSE between adjacent feature map pairs is substantially lower than that between adjacent individual frames. Moreover, unlike the frame-level MSE, the pair-level MSE remains consistently low even as the feature map pair index increases, indicating that the recurring temporal relationship persists across more distant frames. Therefore, feature map pair-level comparison exposes temporal similarity that is obscured by isolated frame-wise analysis and enables more accurate temporal candidate selection for token reduction.

### C. Recovery Opportunity in Video VAR

As discussed in Section III-A, recovering multiple positions from a shared representative can introduce spatial and temporal distortions. This limitation suggests that recovery should preserve the position-specific information of each skipped token rather than propagate features across different locations. The remaining challenge is therefore to identify a suitable representation that enables such recovery accurately.

Video VAR models employ a coarse-to-fine generation scheme. Early scales rapidly construct coarse visual structures,

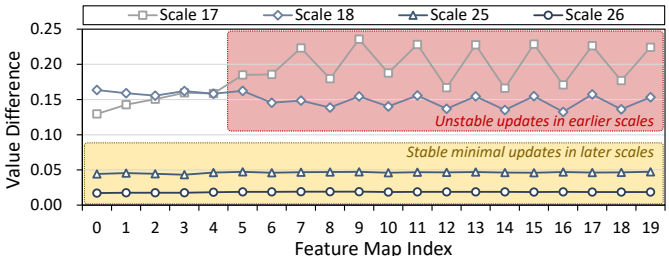



Fig. 7. Average updated absolute value compared to the previous scale across frames in early scales (17, 18) and later scales (25, 26) of InfinityStar-8B.

whereas later scales progressively refine fine-grained details. This suggests that features may change only slightly between adjacent later scales. To examine this property, we analyze the inter-scale variation of value features, as shown in Fig. 7. Specifically, we measure the update magnitude between each current-scale feature and its position-aligned feature from the preceding scale.

At early scales, such as Scales 17 and 18, the updates are large and fluctuate substantially across frames, indicating that the visual representation is still being actively formed. At later scales, such as Scales 25 and 26, the updates remain small and stable across frames, indicating that these scales primarily refine the features generated at the preceding scale. This observation suggests that a skipped position can be recovered using its position-aligned preceding-scale feature without propagating information from a different spatial or temporal position.

## IV. Autobahn Algorithm

Based on these observations, we introduce Autobahn, a semantic–temporal reduction algorithm with previous-scale recovery for video VAR generation. Because video features are progressively refined across scales, similarity between features does not necessarily imply that their updates can be safely skipped. Redundancy identification must therefore account for the semantic and temporal structure of these evolving features to distinguish redundant computation from updates needed for subsequent refinement. To this end, Autobahn performs position-aligned comparisons with the reference image and across adjacent-frame pairs, capturing semantic consistency and interleaved temporal redundancy, respectively, with low comparison overhead. However, skipped positions must be accurately recovered to mitigate the propagation of token reduction errors across subsequent decoding scales. To this end, inspired by FastVAR [4], Autobahn exploits the strong correlation between successive scales to recover these positions using their corresponding previous-scale features, preserving position-specific information with minimal overhead. Fig. 8 summarizes the resulting four-stage pipeline: semantic masking, temporal masking, token update, and token recovery.

### A. Semantic-Temporal Identification

**Reference-guided Semantic Identification.** Semantic masking first exploits the semantically established represen-

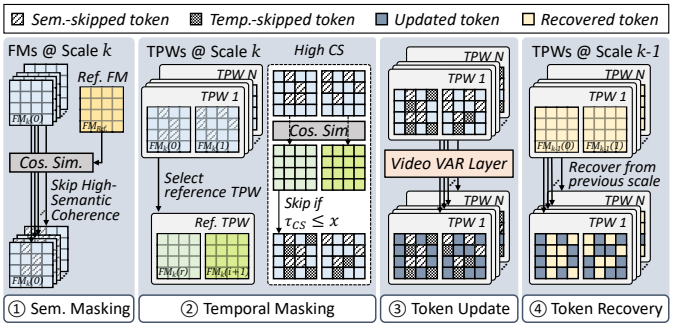



Fig. 8. Overview of the Autobahn Algorithm

tation of the reference image, as shown in stage 1 of Fig. 8. Unlike intermediate video-frame features that are still being progressively refined, the reference image provides a reliable semantic basis for identifying visually unimportant features, such as those corresponding to static background regions or static objects in the foreground.

In detail, given the latent feature at scale $k$, frame $t$, and spatial position $p$, denoted by $x_{k,t,p}$, Autobahn measures its cosine similarity with the position-aligned reference-image feature $r_{k,p}$ as

$$s^{\text{sem.}}_{k,t,p} = \frac{x^{\top}_{k,t,p} r_{k,p}}{\|x_{k,t,p}\|_2 \, \|r_{k,p}\|_2} \tag{1}$$

When $s^{\text{sem.}}_{k,t,p} \geq \tau_{\text{sem.}}$, the feature is classified as semantically stable, since its information is already captured by the fully established semantics of the position-aligned reference image feature. While this position is directly marked as a high-confidence skip position, the positions that do not meet the threshold remain unresolved and are forwarded to the temporal masking stage, where their similarity to other video-frame features is further examined.

The remaining unresolved features are passed to temporal masking, where their redundancy with other video-frame features is further examined. By restricting temporal masking to this smaller set, Autobahn keeps the identification cost proportional to the candidate set rather than the full token count, while avoiding premature removal of potentially important information.

**TPW-based Temporal Identification.** Next, Autobahn examines the remaining positions using temporal redundancy within the video. As discussed in Section III-B, video VAR feature maps do not follow the smooth temporal similarity typically observed between adjacent frames in natural video data. Instead, they exhibit an interleaving pattern that cannot be reliably captured through individual frame-wise comparisons.

To capture the interleaved temporal structure, Autobahn introduces a TPW that groups two adjacent feature maps. $TPW^i_k$ comprises $FM_{2i}$ and $FM_{2i+1}$ as a unit for temporal redundancy identification, where $k$ and $i$ denote the scale and TPW indices, respectively. Each TPW contains $2L$ latent features, with $L$ spatial positions per frame. At each spatial position, the features from the two frames form a pair that is jointly compared with the corresponding pair in a reference TPW to determine whether updates at that position can be skipped.

To faithfully identify the temporally representative TPW, Autobahn selects the center TPW as a shared temporal reference for the entire clip, as shown in stage 2 of Fig. 8. Since the center minimizes the maximum temporal distance to any target TPW, it serves both earlier and later temporal regions evenly. Using a single shared reference also simplifies buffering and eliminates the overhead of selecting a different reference. Based on this criterion, temporal identification is performed on each target TPW only at spatial positions that remain unresolved in the two frames after reference-guided selection. To fully exploit the temporal interleaving pattern, Autobahn concatenates the feature vectors from the two frames to form $w_{k,i,p}$, where $k$, $i$, and $p$ denote the scale, TPW index, and spatial position, respectively. It then computes the similarity between $w_{k,i,p}$ and the representation at the same position in the reference TPW, $w_{k,r,p}$, where $r$ denotes the reference TPW index, as follows:

$$s^{\text{TPW}}_{k,i,p} = \frac{w^{\top}_{k,i,p} w_{k,r,p}}{\|w_{k,i,p}\|_2 \, \|w_{k,r,p}\|_2}, \qquad i \neq r. \tag{2}$$

If $s^{\text{TPW}}_{k,i,p} \geq \tau_{\text{TPW}}$, the unresolved features at position $p$ are classified as temporally redundant and marked for skipping. Otherwise, they are retained for subsequent scale-wise refinement.

Together, the two stages determine the final skip mask, where a set of positions is skipped if it is identified by either the reference-guided stage, $m^{\text{sem.}}_{k,t,p}$, or the TPW-level temporal stage, $m^{\text{TPW}}_{k,t,p}$:

$$m^{\text{skip}}_{k,t,p} = m^{\text{sem.}}_{k,t,p} \vee m^{\text{TPW}}_{k,t,p} \tag{3}$$

Through this two-stage process, Autobahn reduces feature-update computation by skipping semantically unimportant and temporally redundant positions without compromising generation quality.

### *B. Position-aligned Previous Scale-based Recovery*

After identification, the skipped features are removed from Transformer computation, leaving the corresponding positions without updated feature values. These missing features must be recovered because each scale serves as the input to subsequent scales, and inaccurate recovery can propagate errors throughout the remaining generation process. An effective recovery scheme must therefore satisfy two requirements. First, it should preserve generation accuracy by reconstructing each skipped position with a feature that closely approximates its updated representation. Second, it should maintain computational efficiency by minimizing the additional computation and metadata required to recover the skipped positions and restore the original feature-map layout.

**Recovery Accuracy.** From the accuracy perspective, the recovered value should closely approximate the feature that would have been produced by the computational update in

the skipped region. To mitigate recovery error, Autobahn exploits the small inter-scale feature changes observed at later scales, as discussed in Section III-C. After the tokens are updated in the video VAR layer and upscaled to the feature map, Autobahn restores the full feature map layout by utilizing position aligned update. Each updated position receives the newly computed video VAR layer output, whereas each skipped position reuses the feature at the corresponding spatio-temporal position from the previous scale. Specifically, a skipped position in $w_{k,i,p}$ is recovered from the scale-aligned $w_{k-1,i,p}$, rather than from the reference TPW, which is used only to construct the skip mask. Thus, Autobahn exploits the small inter-scale feature map changes at later scales to recover skipped positions from an already generated and structurally aligned representation.

This position-aligned recovery becomes particularly important at high sparsity. In representative-based recovery schemes, reducing the number of retained tokens generally increases the number of skipped positions that require reconstruction based on the remaining representatives. Consequently, the same recovery value is assigned to more spatial or temporal positions, suppressing their position-specific differences and increasing feature-map distortion. Because the recovered feature map is used in subsequent scales, this distortion can propagate through the remaining generation process. In contrast, Autobahn recovers each skipped position from its own position-aligned preceding-scale feature. Increasing sparsity therefore increases the number of reused features, but does not increase the number of positions represented by each recovery source.

**Recovery Complexity.** From the complexity perspective, the Autobahn algorithm focuses on simplifying the recovery path after token compaction. While prior methods require interpolation or inverse mapping to recover multiple positions from a single representative, Autobahn only tracks whether each position is skipped or updated, as shown in stage 4 of Fig. 8. Each skipped position is then restored using the position-aligned feature from the preceding scale. As a result, Autobahn reduces recovery metadata overhead and avoids additional recovery operations, enabling lightweight recovery.

At implementation time, the skip-bit table and prefix-sum results already generated during compaction are directly reused for recovery. As the Recovery Unit traverses the original positions in order, an updated position reads its Transformer output using the compaction-buffer base address and its prefix-sum-derived offset, whereas a skipped position selects the corresponding feature from the preceding scale. Recovery therefore requires only deterministic address generation and a two-way selection, without a separate inverse-mapping table, representative lookup, or interpolation datapath. Moreover, sharing the same metadata between compaction and recovery avoids duplicate bookkeeping and maintains a consistent mapping between the compacted sequence and the original feature layout.

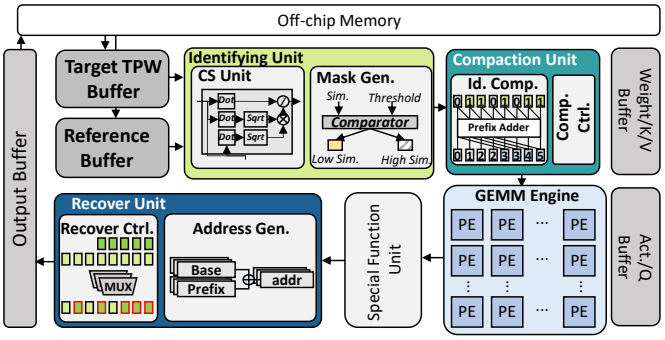



Fig. 9. Overview of the Autobahn Hardware.

## V. Autobahn Hardware Design

Although the Autobahn algorithm reduces identification overhead by identifying skip tokens based on reference features and recovery overhead by reusing previous-scale features, the generated skip mask still introduces irregular data movement in the hardware perspectives. In particular, non-skipped tokens are sparsely distributed across the feature map, preventing them from being directly mapped onto dense computation tiles. Executing this irregular layout leaves many PE lanes idle and incurs fragmented feature accesses, thereby limiting the benefits of token skipping. The non-skipped tokens must therefore be compacted into a contiguous layout before Transformer computation. While compaction deforms the original layout to dense non-skipped tokens, it is critical to recover back to the original layout in generative model. To address these challenges, we propose the Autobahn hardware, a hardware accelerator tailored to mitigate the identification, compaction, and reorganization overhead.

### A. Hardware Overview

As described in Fig. 9, Autobahn hardware is composed of three main components, which are Identifying Unit, Compaction Unit, and Recover Unit. Identifying Unit is a specialized hardware that verifies the redundancy of reduction target and masks the redundant features. It computes the similarity between target and reference features, and generates a skip-bit table indicating which features can bypass Transformer computation. The Compaction Unit uses an index compressor to identify non-skip tokens based on the skip-bit table and gather them into a dense compaction buffer. These operations are coordinated by the Compaction Controller, allowing GEMM and Special Function Units to execute model computation with high MAC unit utilization. After that, the Recover Unit reconstructs the full feature-map layout by combining updated values from the compaction buffer with the previous scale features at the skipped positions. Here, the address generator and a multiplexer select the appropriate value for each output position under the control of the Recovery Controller and write the reordered results to the output buffer.

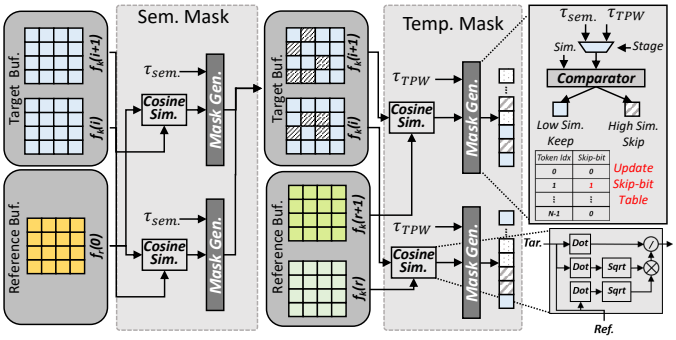



Fig. 10. Detailed architecture of the Identifying Unit.

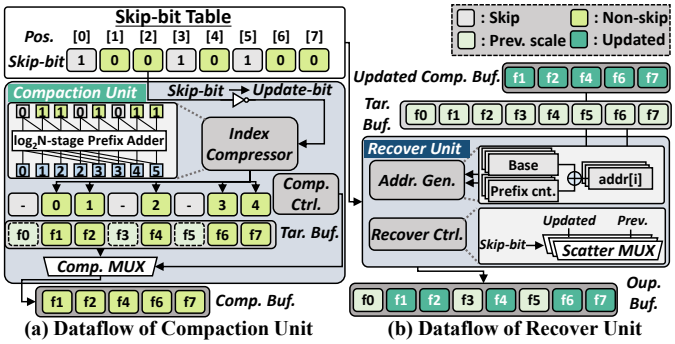



Fig. 11. Reorganization dataflow of Autobahn Hardware.

### B. Detailed Hardware Design

**Identifying Unit.** Fig. 10 illustrates the architectural detail of the Identifying Unit. It has two functionalities which are similarity detection and mask generation. To support this, it is composed of cosine-similarity unit and mask generator. Initially, the cosine-similarity unit receives target and reference image feature from target and reference buffer, respectively. Then, it computes the cosine similarity between the two features using dot product, square root, multiply and division. Notably, similarity is computed only between features at the same spatial position, without considering neighboring positions. As a result, it produces the cosine similarity of the target and reference features in each corresponding positions.

To select the features eligible for skipping, the mask generator receives the semantic cosine similarity and threshold $\tau_{\text{sem.}}$. Its comparator classifies each position as either skippable or requiring an update and records the result in the skip-bit table, where 1 denotes a skippable position and 0 denotes a remaining candidate. After semantic masking is completed, the Identifying Unit performs temporal masking only on positions whose skip bits remain 0. For each position, the stored skip bit controls the enable signal of the temporal cosine engine. If the bit is 1, the position bypasses the temporal similarity computation and its skip bit remains unchanged. Otherwise, the corresponding features are compared with those at the same position in the reference TPW. If the resulting similarity satisfies the temporal threshold $\tau_{\text{temp.}}$, the mask generator updates the corresponding skip bit from 0 to 1; otherwise, it retains the bit as 0. In this manner, the semantic and temporal masking stages progressively update the same skip-bit table, while temporal masking processes only the candidates remaining after semantic masking. This control reuses the existing read and write logic of the skip-bit table without requiring an additional similarity datapath. The resulting skip bits are then forwarded to the Compaction Unit.

**Compaction Unit.** After the Identifying Unit finalizes the skip-bit table, the token layout remains in the original TPW order. Directly feeding this sparse layout to the GEMM engine would lead to underutilization and irregular memory accesses during Transformer computation. The Compaction Unit therefore converts the sparse layout into a dense layout based on the skip-bit table and stores the compacted tokens in the compaction buffer as shown in Fig. 11(a). Specifically, it scans the skip-bit table in the original token order, gathers only the non-skipped tokens from the current-scale feature buffer, and writes them sequentially to the compaction buffer.

To keep compaction hardware-friendly, the Compaction Unit performs mask scanning in fixed-size chunks rather than processing the entire feature map as one long sequence. For each chunk, the Compaction Unit counts the number of updated tokens, assigns local dense indices, and combines them with a chunk offset to generate global dense indices. Since the scan follows the original token order, the Compaction Unit scans the skip-bit table in the original layout order and writes the non-skipped tokens sequentially into the compaction buffer. This property is important for recovery because the Recover Unit can later reconstruct the full feature map using a sequential comparison between the current output position and the next entry of the skip-bit table, rather than relying on associative lookup or random scatter metadata.

The dense token sequence is subsequently processed by the conventional Transformer compute engine, where the GEMM engine executes attention and FFN matrix operations and the Special Function Unit handles normalization, softmax, and activation. Therefore, Autobahn does not require sparse execution support or modification of the existing Transformer datapath.

**Recover Unit.** After Transformer computation, the compacted outputs must be restored to the original token layout and combined with the skipped tokens. As shown in Fig. 11(b), the Recover Unit performs this restoration using two inputs: the updated non-skip features from the compaction buffer and the same-position base features from the target buffer. Because the target buffer already maintains the full-layout feature map produced through the previous-scale refinement path, skipped tokens can be directly retained without an additional bypass feature buffer.

The Recover Unit restores the compacted outputs using an address generator and a recover controller, driven by the skip-bit table and the prefix-sum results from the Index Compressor of Compaction Unit. To restore each token to its original layout position, the unit must determine its destination address and, in case of non-skipped token, location in the compaction buffer.

TABLE I
HARDWARE CONFIGURATION OF AUTOBAHN.

| Hardware Component | Configuration |
|---|---|
| PE Array | $32 \times 32$ PE array<br>BF16 Mul, FP32 Acc |
| Identifying Unit | 4-lane cosine engine<br>32 elements/lane/cycle |
| Compaction Unit | Prefix-sum compaction<br>8 entries/cycle |
| Recover Unit | Position-aligned recovery<br>4 tokens/cycle |

TABLE II
AREA AND POWER BREAKDOWN OF AUTOBAHN.

| Component | Area ($\mathrm{mm}^2$) | Power (W) |
|---|---|---|
| PE Array | 1.03 (26.48%) | 0.390 (60.37%) |
| Identifying Unit | 0.29 ( 7.46%) | 0.047 ( 7.28%) |
| Compaction Unit | 0.02 ( 0.51%) | 0.004 ( 0.62%) |
| Recover Unit | 0.03 ( 0.77%) | 0.009 ( 1.39%) |
| Special Function Unit | 0.10 ( 2.57%) | 0.036 ( 5.57%) |
| Input SRAM | 0.84 (21.59%) | 0.058 ( 8.98%) |
| Weight SRAM | 0.51 (13.11%) | 0.034 ( 5.26%) |
| Output SRAM | 1.00 (25.71%) | 0.063 ( 9.75%) |
| Skip Overlay SRAM | 0.07 ( 1.80%) | 0.005 ( 0.77%) |
| **Total** | **3.89** | **0.646** |

Accordingly, an internal layout counter generates the position aligned addresses for the target buffer and the recovered output buffer. When the skip bit is 0, the address generator adds the corresponding prefix-sum value to the compaction buffer base address to locate the updated Transformer output. The recover controller uses the skip bit as the select signal of a multiplexer. On the other hand, when the skip bit is 1, it retains the same-position previous-scale feature from the target buffer. Buffer reads are gated accordingly to avoid unnecessary accesses. Because the prefix-sum result already represents each non-skipped token's dense index, the updated outputs can be restored directly to their original positions without recomputing prefix counts or maintaining a separate position list.

This recovery dataflow improves both accuracy and hardware efficiency. From the accuracy perspective, skipped positions retain their same-position features rather than being filled with zeros or interpolated from neighboring tokens, thereby avoiding artificial spatial artifacts. From the hardware perspective, the Recover Unit eliminates a skip-position list, an unmerge table, and a separate recovery kernel. The skip-bit table used for compaction is directly reused for recovery, and the full feature map is generated through sequential reads and writes. As a result, sparsity handling is confined to the Identifying, Compaction, and Recover Units, while the main Transformer engine, which is composed of GEMM engine and Special Function Unit, remains dense and regular.

## VI. EVALUATION

### A. Methodology

**Models and Datasets.** We evaluate Autobahn on the 480p and 720p configurations of InfinityStar-8B [15], where each configuration is composed of 28 scales and 30 scales, respectively. We use the prompt suite provided by VBench [6] and EvalCrafter [16] to generate evaluation videos and measure Peak Signal-to-Noise Ratio (PSNR) against the outputs of the original model without token reduction. We evaluate token sparsity levels of 50, 60, 70, and 80% to analyze the trade-off between token reduction and generation quality across different resolutions.

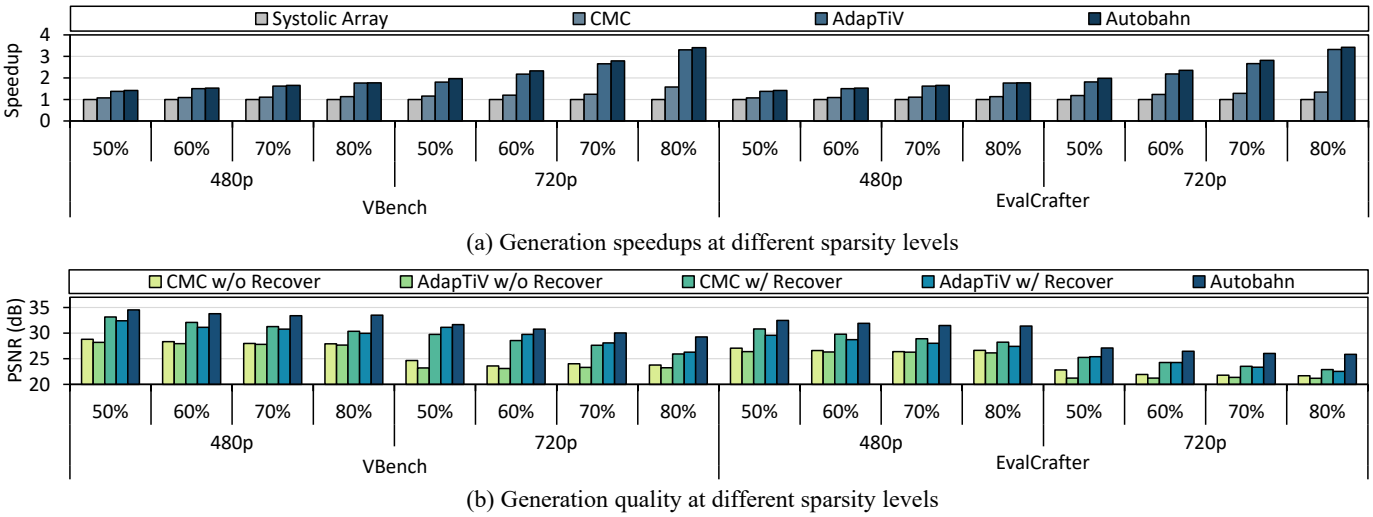
**Algorithmic Baselines.** We compare Autobahn against two state-of-the-art token-reduction methods, CMC [21] and AdapTiV [32]. Since these methods were not originally designed for video VARs, we adapt their token identification and recovery policies to the InfinityStar generation pipeline. CMC [21] identifies redundant tokens using inter-frame similarity and recovers the reduced regions using informative tokens. AdapTiV [32] identifies redundant tokens through intra-frame local matching but does not provide a dedicated recovery mechanism. We therefore restore its reduced tokens through an unmerge operation. All methods are executed on the NVIDIA RTX A6000 GPU using BF16 precision and evaluated under matched sparsity levels to ensure a fair comparison of generation quality. Furthermore, we start applying the reduction methods at the same scale to measure the effectiveness of each method.

**Hardware Configurations and Implementations.** Our hardware configuration is summarized in Table I. To evaluate architectural performance, we develop a cycle-accurate simulation framework that models Autobahn's feature identification, compaction, Transformer execution, and recovery. The simulator consumes scale-wise traces exported from our PyTorch InfinityStar implementation, including feature dimensions, sparsity patterns, and active-feature indices for each evaluated prompt. These traces enable cycle-level accounting of the full pipeline.

We compare Autobahn against CMC [21] and AdapTiV [32] at a common operating frequency of 500 MHz. For a fair iso-area comparison, we scale all accelerator baselines using a consistent unit-area model to match the same total chip area. All designs are configured with 1 MB of SRAM and 64 GB/s of DRAM bandwidth. For area and power evaluation, we synthesize the core logic with Synopsys Design Compiler using the FreePDK 45 nm library [22] and estimate the area and energy of on-chip SRAM buffers with CACTI [18].

### B. Performance and Quality Evaluation

To evaluate the speedup and hardware efficiency, we compare Autobahn against the systolic array, AdapTiV, and CMC on InfinityStar-8B at 480p and 720p across diverse sparsity ranging from 50% to 80%. All speedups and energy efficiencies are normalized to the systolic array baseline without token reduction.




(a) Generation speedups at different sparsity levels




(b) Generation quality at different sparsity levels

Fig. 12. Comparison of video (a) generation speedup at different sparsity levels and (b) generation quality. Speedup is normalized to the Systolic Array.

**Performance.** Autobahn consistently achieves the highest performance across all evaluated sparsity levels, with its average speedup increasing from $1.67\times$ at 50% sparsity to $2.46\times$ at 80%. In comparison, CMC improves only from $1.11\times$ to $1.34\times$. Although all methods reduce Transformer computation through token reduction, broader candidate search of CMC requires numerous feature comparisons and accesses during identification. This overhead offsets a substantial portion of the saved Transformer computation, limiting its performance improvement.

AdapTiV achieves considerably higher speedups than CMC, ranging from $1.58\times$ to $2.41\times$, because its local search space substantially reduces identification overhead. Autobahn further restricts identification through reference-guided masking and TPW-level comparison in a position-aligned manner. Consequently, Autobahn slightly but consistently outperforms AdapTiV across the evaluated sparsity levels.

Autobahn's performance benefit becomes more pronounced at higher resolution. At 80% sparsity, its speedup increases from $1.77\times$ at 480p to $3.41\times$ at 720p, compared with $1.13\times$ and $1.58\times$ for CMC and $1.76\times$ and $3.31\times$ for AdapTiV, respectively. Since the larger Transformer workload at 720p provides greater computation savings, Autobahn's lightweight, linear identification and recovery overhead is more effectively amortized.

**Generative Quality.** Fig. 12(b) compares the generative quality of CMC, AdapTiV, and Autobahn under different sparsity levels. We measure PSNR against videos generated by the original InfinityStar model without token reduction. Therefore, a higher PSNR indicates that a method more faithfully preserves the output of the dense model.

For CMC [21] and AdapTiV [32], we additionally evaluate variants without recovery to quantify the importance of reconstructing the reduced positions. At 80% sparsity, recovery improves the PSNR of CMC [21] by 2.43 dB and 2.13 dB at 480p and 720p, respectively. Similarly, AdapTiV [32] obtains improvements of 2.31 dB and 3.03 dB. These results demonstrate that recovering the reduced regions is essential for maintaining generation quality.

Autobahn consistently achieves the highest PSNR across all evaluated resolutions and sparsity levels. At 80% sparsity, Autobahn achieves 33.49 dB at 480p and 29.26 dB at 720p. In comparison, CMC [21] with recovery achieves 30.34 dB and 25.91 dB, while AdapTiV [32] with recovery achieves 29.97 dB and 26.28 dB. Thus, Autobahn outperforms CMC [21] by 3.15 dB and 3.35 dB and AdapTiV [32] by 3.52 dB and 2.98 dB at 480p and 720p, respectively. Moreover, when sparsity increases from 50% to 80%, Autobahn's PSNR decreases by only 1.03 dB at 480p and 2.41 dB at 720p, while they decrease substantially for CMC [21] and AdapTiV. This smaller quality degradation indicates that Autobahn remains more robust under aggressive token reduction, consistent with its use of reference-guided identification, TPW-level temporal checking, and spatially aligned previous-scale recovery.

### C. Area, Power and Energy

**Area and Power.** To quantify Autobahn's hardware overhead, Table II provides a detailed breakdown of its area and power consumption. The Identifying, Compaction, and Recover Units occupy 7.46%, 0.51%, and 0.77% of Autobahn's total 3.89 $mm^2$ area, respectively, while accounting for 7.28%, 0.62%, and 1.39% of its total 0.646 W power consumption. Together, these units account for only 8.74% of the total area and 9.29% of the total power, demonstrating that Autobahn improves sparse execution efficiency with modest hardware cost. Among the three dedicated units, the Identifying Unit incurs the largest overhead because it performs cosine-similarity computations for token identification. In contrast, the Compaction and Recover Units rely primarily on lightweight prefix-sum and address-generation logic, resulting in negligible area and power overheads. Most of the hardware cost therefore remains concentrated in the PE array and on-chip SRAMs

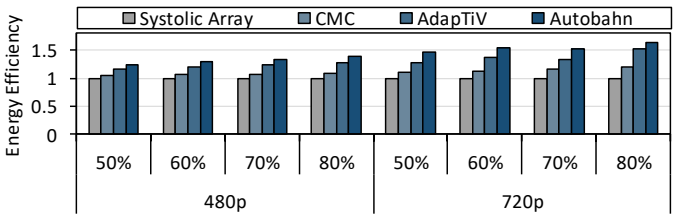



Fig. 13. Energy efficiency at different sparsity levels.

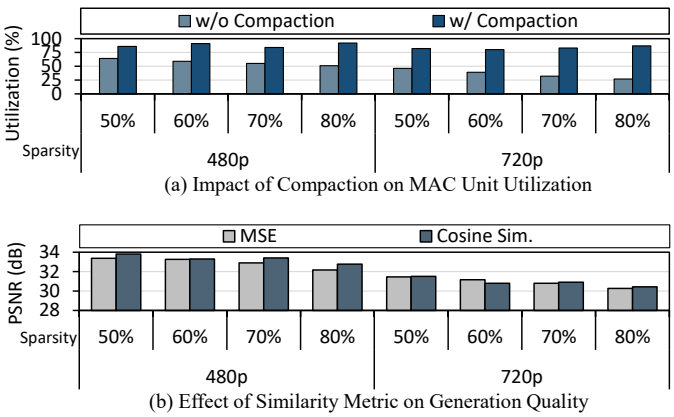
rather than in the additional logic required to support sparse execution.

**Energy Efficiency.** We evaluate the energy efficiency of Autobahn and the hardware baselines, as shown in Fig. 13. Across all resolutions and sparsity levels, Autobahn achieves an average energy efficiency of $1.42\times$, compared with $1.29\times$ for AdapTiV [32] and $1.11\times$ for CMC. This efficiency trend primarily reflects the size of the candidate search space used for token identification: CMC performs a segment-level search, whereas AdapTiV restricts its comparisons to a local neighborhood. Autobahn further confines identification to position-aligned candidates selected through reference-guided pruning, substantially reducing the number of similarity computations. This restricted search space also decreases feature accesses and movement during identification, further lowering its energy overhead. Consequently, a larger fraction of the energy saved by token reduction translates into end-to-end efficiency gains, allowing Autobahn to consistently outperform both baselines.

### *D. Design Space Exploration*

In this section, we explore Autobahn's architectural and algorithmic design choices. First, we evaluate how the Compaction Unit improves MAC utilization by reorganizing irregularly retained tokens into dense tiles. As shown in Fig. 14(a), the irregular token layout without compaction leads to substantial MAC underutilization, particularly at high sparsity. As sparsity increases, MAC utilization decreases from 64% to 51% at 480p and from 46% to 27% at 720p, corresponding to an average relative reduction of 30.8%. In contrast, the Compaction Unit reorganizes the retained tokens into a dense layout, maintaining stable MAC utilization of approximately 91% across all sparsity levels. This dense organization minimizes inactive MAC operations caused by irregular token placement and partially filled execution tiles. Consequently, the Compaction Unit allows the computational benefits of increased sparsity to translate into proportional execution-time reductions without sacrificing array utilization.

We further investigate the effect of the similarity metric used for token identification on generation quality, as shown in Fig. 14(b). MSE directly captures differences in feature magnitude with lower computational complexity, whereas cosine similarity measures directional similarity independently of feature magnitude, enabling more reliable identification of semantically redundant tokens. At 80% sparsity, cosine similarity improves generation quality by 0.6 dB over MSE. Although this improvement is moderate, it becomes important under




(a) Impact of Compaction on MAC Unit Utilization

(b) Effect of Similarity Metric on Generation Quality

Fig. 14. (a) Impact of the Compaction Unit on MAC utilization. (b) Effect of the similarity metric on generation quality.

aggressive token reduction, where identification errors can propagate through subsequent generation scales. Moreover, Autobahn's restricted position-aligned search space limits the number of similarity computations, mitigating the additional cost of cosine normalization. Autobahn therefore adopts cosine similarity to prioritize reliable token identification and preserve generation quality.

## VII. Related Works

Token reduction has been extensively explored for Vision Transformers [1], [9], [13], [20], [31] and their hardware accelerators [3], [7], [32], [33], and has been extended to video models by exploiting spatial and temporal similarity [2], [12], [14], [21], [36]. These works share a common premise that the visual content is already formed, but Video VARs violate this premise. The representation is progressively constructed across decoding scales, the temporal structure of the generated features does not follow that of natural video, and every removed position must be reconstructed with sufficient fidelity to condition the following scales.

For identification, prior methods determine redundancy from the similarity among already-available features, either within a local spatial neighborhood [1], [19], [32] or between a representative frame and the remaining frames of a temporal segment [2], [21], [28]. Autobahn instead recognizes the intermediate reference image of video VAR as a semantic cue for reduction. For the remaining positions, Autobahn further observes an interleaving temporal pattern that frame-wise omparison [14], [17], [36] cannot capture, and evaluates temporal redundancy at the granularity of a Temporal Probing Window.

For recovery, prior methods either copy a shared representative back to all merged positions [1], [9], [32], propagate the features of a representative frame to skipped frames [21], or interpolate removed tokens from their neighbors [26], [34]. In every case a single source value is assigned to multiple spatially or temporally distinct positions, which suppresses position-specific detail and allows the error to propagate through subsequent decoding scales. By reusing the feature

already available at the preceding scale for each skipped position, Autobahn improves generation quality with minimal error.

## VIII. Conclusion

Video VAR has emerged as a promising video generation model, yet the rapidly growing spatio-temporal token sequences and the high overhead of redundancy identification pose substantial computational bottlenecks. We present Autobahn, a semantic–temporal token reduction algorithm and hardware co-design for efficient video VAR generation. We further observe that the coarse-to-fine hierarchy preserves position-aligned information across consecutive scales, enabling lightweight recovery from preceding-scale features. Based on these observations, Autobahn combines reference-guided semantic pruning, TPW-based temporal identification, and previous-scale recovery with specialized hardware that compacts retained tokens into a dense layout for efficient MAC execution. Autobahn achieves up to $2.54\times$ speedup and improves PSNR by an average of $3.8\,\mathrm{dB}$ over the baseline accelerators.

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